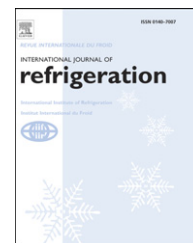


available at [www.sciencedirect.com](http://www.sciencedirect.com)journal homepage: [www.elsevier.com/locate/ijrefrig](http://www.elsevier.com/locate/ijrefrig)

# Dynamic operation of an active magnetic regenerator (AMR): Numerical optimization of a packed-bed AMR

Jaka Tušek\*, Andrej Kitanovski, Ivan Prebil, Alojz Poredoš

University of Ljubljana, Faculty of Mechanical Engineering, Aškerčeva c. 6, 1000 Ljubljana, Slovenia

## ARTICLE INFO

### Article history:

Received 10 February 2011

Received in revised form

13 April 2011

Accepted 20 April 2011

Available online 1 May 2011

### Keywords:

Magnetic refrigerator

Regenerator

Modeling

Simulation

Optimization

Gadolinium

## ABSTRACT

A new, fast and flexible, time-dependent, one-dimensional numerical model was developed in order to study in detail the operation of an active magnetic regenerator (AMR). The model is based on a coupled system of equations (for the magnetocaloric material and the heat-transfer fluid) that have been solved simultaneously with the software package MATLAB. The model can be employed to analyze a wide range of different operating conditions (mass-flow rate, operating frequency, magnetic field change), different AMR geometries, different magnetocaloric materials and heat-transfer fluids, layered and single-bed AMRs, etc.

This paper also presents an optimization of the AMR's geometry, where the AMR consists of a packed-bed of grains (spheres) of gadolinium (Gd). The optimization of the mass-flow rate and the operating frequency of the AMR were performed by studying five different diameters of Gd spheres.

© 2011 Elsevier Ltd and IIR. All rights reserved.

# Fonctionnement dynamique d'un régénérateur actif magnétique (AMR) : optimisation numérique d'un AMR à lit fixe rempli

Mots clés : réfrigérateur ; régénérateur ; modélisation ; simulation ; optimisation ; gadolinium

## 1. Introduction

Magnetic refrigeration is a technology that exploits a physical phenomenon known as the magnetocaloric effect (MCE). This effect causes magnetocaloric materials to heat up when inserted into a magnetic field and to cool down when removed from the field. With the proper technology the MCE can be

used in a cyclical system to obtain a continuously operating magnetic refrigerator. The technology that has so far proven to be the most promising is magnetic refrigeration based on an active magnetic regenerator (AMR). This AMR has a porous structure and represents the "heart" of the magnetic refrigeration device. It has the role of the refrigerant (it contains the magnetocaloric material) as well as operating as a

\* Corresponding author. Tel.: +386 1 4771 418; fax: +386 1 2518 567.

E-mail address: [jaka.tusek@fs.uni-lj.si](mailto:jaka.tusek@fs.uni-lj.si) (J. Tušek).

0140-7007/\$ – see front matter © 2011 Elsevier Ltd and IIR. All rights reserved.

doi:10.1016/j.ijrefrig.2011.04.007

**Nomenclature***Variables*

|               |                                                               |
|---------------|---------------------------------------------------------------|
| $A_{ht}$      | heat-transfer area [m <sup>2</sup> ]                          |
| $A_o$         | frontal area [m <sup>2</sup> ]                                |
| $\alpha$      | heat-transfer coefficient [Wm <sup>-2</sup> K <sup>-1</sup> ] |
| $c$           | specific heat [Jkg <sup>-1</sup> K <sup>-1</sup> ]            |
| $B$           | internal magnetic field [T]                                   |
| COP           | coefficient of performance                                    |
| $d_h$         | hydraulic diameter [m]                                        |
| $d$           | sphere diameter [m]                                           |
| $\varepsilon$ | porosity                                                      |
| $f_o$         | friction factor                                               |
| $\lambda$     | thermal conductivity [Wm <sup>-1</sup> K <sup>-1</sup> ]      |
| $L$           | length [m]                                                    |
| $\dot{m}$     | mass-flow rate [kgs <sup>-1</sup> ]                           |
| $m$           | mass [kg]                                                     |
| $Nu$          | Nusselt number                                                |
| $\eta$        | pump efficiency                                               |
| $\nu$         | frequency [Hz]                                                |
| $P$           | fluid flow period [s]                                         |
| $Pr$          | Prandtl number                                                |
| $\Delta p$    | pressure drop [Pa]                                            |
| $\dot{Q}$     | power/load [W]                                                |
| $Re$          | Reynolds number                                               |
| $\rho$        | density [kgm <sup>-3</sup> ]                                  |

|                 |                                                       |
|-----------------|-------------------------------------------------------|
| $s$             | specific entropy [Jkg <sup>-1</sup> K <sup>-1</sup> ] |
| $T$             | temperature [K]                                       |
| $\Delta T_{ad}$ | adiabatic temperature change [K]                      |
| $t$             | time [s]                                              |
| $\Delta t$      | time step [s]                                         |
| $\tau$          | dwel time [s]                                         |
| $v$             | velocity [ms <sup>-1</sup> ]                          |
| $V_{AMR}$       | AMR volume [m <sup>3</sup> ]                          |
| $W$             | work input [W]                                        |
| $V^*$           | ratio of displaced fluid mass                         |
| $\Delta x$      | spatial step [m]                                      |

*Indices*

|       |                     |
|-------|---------------------|
| $c$   | cold                |
| $C$   | Curie               |
| CHEx  | cold heat exchanger |
| $e$   | electronic          |
| $eff$ | effective           |
| $f$   | fluid               |
| $fin$ | final               |
| $h$   | hot                 |
| HHEX  | hot heat exchanger  |
| $i$   | initial             |
| $in$  | inlet/input         |
| $l$   | lattice             |
| $m$   | magnetic            |
| $s$   | solid               |

regenerator. The heat regeneration between the particular phases of the cooling cycle leads to an increased temperature span. This is required because the magnetocaloric effect of current magnetocaloric materials is severely limited by the temperature interval in which a large magnetocaloric effect occurs. The AMR usually performs an inverse (left thermodynamic) Brayton refrigeration cycle (at each differential point along its length) and thus has the following four phases: 1) adiabatic magnetization, 2) isofield heat-transfer (fluid flow from the cold to the hot external heat exchanger), 3) adiabatic demagnetization, and 4) isofield heat-transfer (fluid flow from the hot to the cold external heat exchanger).

In order to understand the operation of the AMR in detail, a number of AMR numerical models have been developed so far. These relate to 1D models which assume that the fluid flow and the thermal conduction (if it is included) occur only in the downstream fluid direction (Smaili and Chahine, 1998; Siddikov et al., 2005; Shir et al., 2005; Engelbrecht et al., 2005; Kawanami, 2007; Šarlah, 2008; Boucekara et al., 2008 and Tagliafico et al., 2009). In general, 1D models neglect the intra-particle thermal conduction and require the application of a heat-transfer coefficient between the fluid and the solid matrix. Recently, 2D models of the AMR were developed by Petersen et al. (2008), Nielsen et al. (2009), Oliveira et al. (2009) and Liu and Yu (2011). They took into account the 2D velocity field, which is established in the downstream direction of the fluid flow. In general, they also considered the thermal conduction in 2D (perpendicular and parallel to the fluid flow). Equations for the fluid and the solid matrix in 2D models are not directly coupled through the heat-transfer coefficient, but

are connected through an additional boundary condition that defines the thermal contact between the fluid and the solid. This is physically more appropriate than the use of the heat-transfer coefficient. This is particularly so because in this case the result does not depend on the accuracy of the correlation for the heat-transfer coefficient. However, due to a random particle distribution in the packed-bed of the AMR, 2D models can only be applied for AMRs with ordered structures (flat parallel plates, square channel, etc.). In general, a fully developed 3D model could be applied to any geometry. The first attempt of the 3D AMR model is presented in Bouchard et al. (2009). Models can be further distinguished with regard to the inclusion of the magnetocaloric effect. Some authors applied an adiabatic temperature change ( $\Delta T_{ad}(T, B_i, B_{fin})$ ) of the magnetocaloric material during the process of magnetization and demagnetization. Another possibility is to include the magnetocaloric effect directly into the mathematical model through  $(\partial M / \partial T)$ , as presented by Shir et al. (2005), or through  $(\partial s / \partial \mu_0 H)$ , as presented by Engelbrecht et al. (2005). A detailed review of all the so far published AMR numerical models is presented in Nielsen et al. (2011).

## 2. Numerical modeling

### 2.1. Mathematical model

The numerical model of the AMR is based on a mathematical model of the heat-transfer in the porous structure of a passive heat regenerator. A set of coupled governing equations for the

fluid (Eq. (1)) and the solid matrix (Eq. (2)) can be derived from the equation for a change of the internal energy. The governing equations are written for the one-dimensional case ( $x$ -direction is defined to be parallel to the fluid flow) and contains the following terms: (I.) the rate of temperature change; (II.) the friction losses due to the pressure drop; (III.) the convective heat-transfer between the fluid and the solid and (IV.) the thermal conduction within the fluid and the solid parallel to the fluid flow. Taking into account the assumptions normally used for a passive heat regenerator analysis (Shah and Sekulic (2003)), the governing equations in the one-dimensional case can be written as follows:

$$\underbrace{\frac{\partial T_f}{\partial t} + v_x \cdot \frac{\partial T_f}{\partial x}}_I = \underbrace{\left[ f_0 \cdot \frac{2 \cdot v_x^3}{d_h \cdot c_f} \right]}_{II} + \underbrace{\frac{\alpha_{eff} \cdot A_{ht}}{c_f \cdot \dot{m}_f \cdot \tau} \cdot (T_s - T_f)}_{III} \quad (1)$$

$$\underbrace{\frac{\partial T_s}{\partial t}}_I = \underbrace{\frac{\lambda_{eff}}{\rho_s \cdot c_s(T, B) \cdot (1 - \varepsilon)} \cdot \frac{\partial^2 T_s}{\partial x^2}}_{IV} + \underbrace{\frac{\alpha_{eff} \cdot A_{ht}}{c_s(T, B) \cdot \dot{m}_s} \cdot (T_f - T_s)}_{III} \quad (2)$$

where the velocity ( $v_x$ ) is defined as:

$$v_x = \frac{\dot{m}}{A_0 \cdot \varepsilon \cdot \rho_f} \quad (3)$$

The thermal conduction within the solid and the fluid are combined and included in the equation for the solid as  $\lambda_{eff}$ , like it was presented by Engelbrecht et al. (2005) and Šarlah (2008). Such an approach simplifies the equation for the fluid and also the numerical solution. Because the one-dimensional model neglects the temperature distribution in the solid perpendicular to the fluid flow the correction factor for the heat-transfer coefficient needs to be applied (the effective heat-transfer coefficient ( $\alpha_{eff}$ )). Using such an approach the effect of the non-uniform temperature distribution is, to a certain extent, taken into account (Engelbrecht et al., 2005). Correlations for the friction factor and the heat-transfer coefficient (Nusselt number) in a spherical, packed-bed regenerator were given by Kays and London (1984) and Wakao and Kegu (1982), respectively:

$$f_0 = 23,462 \cdot \text{Re}^{-0.6716} \quad (4)$$

$$\text{Nu} = 2 + 1,1 \cdot \text{Re}^{0.6} \cdot \text{Pr}^{\frac{1}{3}} \quad (5)$$

In order to obtain a fully defined system of equations the initial and boundary conditions have to be defined. The required initial and boundary conditions are: the fluid inlet temperature is pre-defined; the solid and fluid temperatures at the end of one phase of the process are the same as at the beginning of the next phase, and an adiabatic boundary condition at both ends of the AMR is assumed:

$$T_f(x=0, t) = T_{f, \text{in}}; \quad T_{s,f}(x, t=P) = T_{s,f}(x, t=0);$$

$$\left. \frac{\partial T_{s,f}}{\partial x} \right|_{x=0} = \left. \frac{\partial T_{s,f}}{\partial x} \right|_{x=L} = 0 \quad (6)$$

The following assumptions were considered in the mathematical model: the fluid is incompressible; no pressure oscillations occur; heat losses to/from the surroundings are negligible; the velocity and temperature of the fluid at the inlet are uniform over the fluid flow cross-section and constant

over time; the heat-transfer coefficient is constant throughout the AMR; the solid in the AMR is uniformly distributed; the heat-transfer due to radiation is negligible; and no fluid carryover leakage occurs. The numerical model, however, comprises three additional assumptions:

- a discrete “on-off” magnetic field change is used and the magnetization and demagnetization processes occur instantaneously;
- an ideal valve system is assumed, so the change in the inlet mass-flow rate is a discrete “on-off” function;
- the external heat exchangers are considered to be ideal.

## 2.2. Inclusion of the magnetocaloric effect

The governing equations presented above do not include the magnetocaloric effect and are thus only suitable for a simulation of the heat-transfer in an isofield process. In order to obtain a complete AMR model the magnetocaloric effect must be included. The simplest and the most straightforward way to do this is to apply the adiabatic temperature change of the magnetocaloric material during the process of magnetization and demagnetization. The values of the adiabatic temperature change can be taken from the appropriate  $\Delta T_{ad}(T, B_i, B_{fin})$  diagrams. These can be found in the literature for different magnetocaloric materials or can be obtained using the mean field theory (MFT), as proposed by e.g. Hashimoto et al. (1981). The temperature change of the magnetocaloric material during the magnetization and demagnetization process is applied as:

$$T_s = T_{s,i} \pm \Delta T_{ad}(T_{s,i}, B_i, B_{fin}) \quad (7)$$

Both magnetocaloric properties for Gd ( $\Delta T_{ad}(T, B_i, B_{fin})$  and  $c_s(T, B)$ ) were obtained with the MFT, together with the combination of experimentally obtained data, in order to get more realistic results. For the purpose of the analysis in this article, the  $c_s(T, B)$  was defined by taking the values of the experimental data presented by Dankov et al. (1998). The experimentally obtained values for the specific heat at 0 and 2 T were used to obtain the specific heat for the magnetic fields between both values. To do so, the ratios of the isothermal entropy change ( $\Delta s(T, B_i, B_{fin})$ ) were applied; they were calculated with the MFT. Such a procedure is based on the fact that the specific heat of the material in the non-zero magnetic field mainly depends on the magnetic entropy change ( $\Delta s_m(T, B_i, B_{fin})$ ), since the contributions of the total entropy of the material ( $s_m(T, 0)$ ,  $s_l(T)$  and  $s_e(T)$ ) are field independent. The adiabatic temperature change of the Gd sample calculated using the MFT shows a weak agreement with the experimentally obtained data. This is particularly the case for temperatures below the Curie point ( $T_C$ ), as also noted by Hashimoto et al. (1981). For this reason, the adiabatic temperature change was corrected using the experimental results of Kawanami (2007). Above described procedure was presented by Šarlah (2008). It should be further noted that the magnetocaloric properties obtained experimentally or corrected using experimental data will lead to a lower performance of the AMR as compared to the properties obtained directly with the MFT (Nielsen et al., 2009). Special attention

should be paid to matching of the curve which describes the magnetocaloric effect (in this case  $\Delta T_{ad}(T, B_i, B_{fin})$ ) and the curve of the specific heat ( $c_s(T, B)$ ). Since only a relatively minimal mismatch between both curves may cause quite significant variations in the AMR's performance characteristics, especially for the COP values calculated by Eq. (14), the safest method is to use the magnetocaloric properties obtained directly by the MFT in order to get thermodynamically consistent results for all operating conditions. However, the correction or any mismatch of the magnetocaloric properties may change the absolute values of the AMR's performance (mostly the COP), but this does not affect the trends of dependency presented in this paper, as they mainly depend on the thermo-hydraulic properties.

Another way to conduct the numerical analyses is to introduce a new mathematical model based on the enthalpy rather than the specific heat (see, e.g., [Egolf and Manz, 1994](#)). However, this issue is left for future work and publications.

### 2.3. Numerical solution

The discretization of the governing equations and the boundary conditions was performed using the finite-difference method, where the implicit Euler method was used for the discretization of time-dependent derivatives. The numerical mesh of the AMR is presented in the [Fig. 1](#). Equations (1) and (2) can be written in the discretized form:

$$\begin{aligned} -K_2 \cdot T_f(i-1, j+1) + \underbrace{(1+K_1)}_{K_{1-1}} \cdot T_f(i, j+1) + K_2 \cdot T_f(i+1, j+1) \\ - K_1 \cdot T_s(i, j+1) = T_f(i, j) + |K_p| \end{aligned} \quad (8)$$

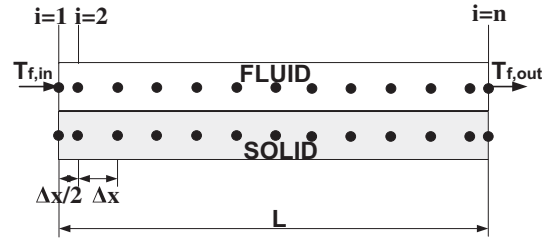


Fig. 1 – One-dimensional AMR numerical mesh.

$$\begin{aligned} -K_4 \cdot T_s(i-1, j+1) + \underbrace{(1+2 \cdot K_4 + K_3)}_{K_{2-1}} \cdot T_s(i, j+1) \\ - K_4 \cdot T_s(i+1, j+1) - K_3 \cdot T_f(i, j+1) = T_s(i, j) \end{aligned} \quad (9)$$

where the coefficients K are defined as:

$$\begin{aligned} K_1 = \frac{\alpha_{eff} \cdot A_{ht} \cdot \Delta t}{\dot{m}_f \cdot c_f \cdot \tau}; \quad K_2 = \frac{L \cdot \Delta t}{2 \cdot \Delta x \cdot \tau}; \quad K_3 = \frac{\alpha_{eff} \cdot A_{ht} \cdot \Delta t}{m_s \cdot c_s(T_s, B)}; \\ K_4 = \frac{\lambda_{eff} \cdot \Delta t}{\rho_s \cdot c_s(T_s, B) \cdot (1-\varepsilon) \cdot \Delta x^2}; \quad K_p = f \cdot \frac{2 \cdot \left(\frac{L}{\tau}\right)^3 \cdot \Delta t}{d_h \cdot c_f} \end{aligned} \quad (10)$$

The discretized boundary conditions take the following form:

$$T_f(i=1, j) = T_{f,in} \quad (11a)$$

$$T_f(i=n-2, j+1) - 4 \cdot T_f(i=n-1, j+1) + 3 \cdot T_f(i=n, j+1) = 0 \quad (11b)$$

$$T_s(i=n-2, j+1) - 4 \cdot T_s(i=n-1, j+1) + 3 \cdot T_s(i=n, j+1) = 0 \quad (11c)$$

|        | coefficients of the fluid equation |      |      |    |   |   |     |      |    |     | coefficients of the solid equation |      |     |   |   |     |      |              |                 |                 |  |  |
|--------|------------------------------------|------|------|----|---|---|-----|------|----|-----|------------------------------------|------|-----|---|---|-----|------|--------------|-----------------|-----------------|--|--|
| Eq. 8a | 1                                  | 0    | 0    | 0  | . | . | .   | .    | 0  | 0   | 0                                  | 0    | .   | . | . | .   | 0    | $T_f(1,j+1)$ | $T_{f,in}$      |                 |  |  |
|        | -K2                                | K1_1 | K2   | 0  | . | . | .   | .    | 0  | 0   | -K1                                | 0    | 0   | . | . | .   | 0    | $T_f(2,j+1)$ | $T_f(2,j)+Kp$   |                 |  |  |
|        | 0                                  | -K2  | K1_1 | K2 | . | . | .   | .    | 0  | 0   | 0                                  | -K1  | 0   | . | . | .   | 0    | $T_f(3,j+1)$ | $T_f(3,j)+Kp$   |                 |  |  |
|        | .                                  | .    | .    | .  | . | . | .   | .    | .  | .   | .                                  | .    | .   | . | . | .   | .    | .            | .               |                 |  |  |
| Eq. 5  | .                                  | .    | .    | .  | . | . | .   | .    | .  | .   | .                                  | .    | .   | . | . | .   | .    | .            | .               |                 |  |  |
|        | .                                  | .    | .    | .  | . | . | .   | .    | .  | .   | .                                  | .    | .   | . | . | .   | .    | .            | .               |                 |  |  |
|        | .                                  | .    | .    | .  | . | . | .   | .    | .  | .   | .                                  | .    | .   | . | . | .   | .    | .            | .               |                 |  |  |
|        | .                                  | .    | .    | .  | . | . | .   | .    | .  | .   | .                                  | .    | .   | . | . | .   | .    | .            | .               |                 |  |  |
| Eq. 8b | 0                                  | 0    | 0    | 0  | . | . | -K2 | K1_1 | K2 | 0   | 0                                  | 0    | 0   | . | . | 0   | -K1  | 0            | $T_f(n-1,j+1)$  | $T_f(n-1,j)+Kp$ |  |  |
|        | 0                                  | 0    | 0    | 0  | . | . | 1   | -4   | 3  | 0   | 0                                  | 0    | 0   | . | . | 0   | 0    | -K1          | $T_f(n,j+1)$    | $T_f(n,j)$      |  |  |
| Eq. 8d | 0                                  | 0    | 0    | 0  | . | . | .   | .    | 0  | -3  | 4                                  | -1   | 0   | . | . | .   | .    | 0            | $T_s(n+1,j+1)$  | 0               |  |  |
|        | 0                                  | -K3  | 0    | 0  | . | . | .   | .    | 0  | -K4 | K2_1                               | -K4  | 0   | . | . | .   | .    | 0            | $T_s(n+2,j+1)$  | $T_s(n+2,j)$    |  |  |
| Eq. 6  | 0                                  | 0    | -K3  | 0  | . | . | .   | .    | 0  | 0   | -K4                                | K2_1 | -K4 | . | . | .   | .    | 0            | $T_s(n+3,j+1)$  | $T_s(n+3,j)$    |  |  |
|        | .                                  | .    | .    | .  | . | . | .   | .    | .  | .   | .                                  | .    | .   | . | . | .   | .    | .            | .               | .               |  |  |
|        | .                                  | .    | .    | .  | . | . | .   | .    | .  | .   | .                                  | .    | .   | . | . | .   | .    | .            | .               | .               |  |  |
|        | .                                  | .    | .    | .  | . | . | .   | .    | .  | .   | .                                  | .    | .   | . | . | .   | .    | .            | .               | .               |  |  |
| Eq. 8c | 0                                  | 0    | 0    | 0  | . | . | 0   | -K3  | 0  | 0   | 0                                  | 0    | 0   | . | 0 | -K4 | K2_1 | -K4          | $T_s(2n-1,j+1)$ | $T_s(2n-1,j)$   |  |  |
|        | 0                                  | 0    | 0    | 0  | . | . | 0   | 0    | 0  | 0   | 0                                  | 0    | 0   | . | 0 | 1   | -4   | 3            | $T_s(2n,j+1)$   | 0               |  |  |

Fig. 2 – System of equations written in a matrix equation.

$$-T_s(i=3,j+1)+4\cdot T_s(i=2,j+1)-3\cdot T_s(i=1,j+1)=0 \quad (11d)$$

A system of algebraic equations is obtained (Eqs. 8–11). The solution of this system was carried out simultaneously with the inverse matrix method, where both equations (for the fluid and the solid) are considered as being part of a single system. The equations can be further rewritten in the matrix equation:

$$[A] \cdot [\vec{T}] = [\vec{B}] \quad (12)$$

where  $A$  is a tridiagonal matrix of the coefficients ( $K$ ),  $T$  is a vector of the new temperatures and  $B$  is a vector of known values (the temperatures of the previous time step, the boundary conditions and the coefficient which describes the viscous losses). Fig. 2 represents an extended form of Eq. (12) with the inclusion of Eqs. (8–11).

However, the matrix equation was further implemented in the MATLAB software package and solved for each time step during each cycle until the steady state is achieved. The flow chart of the program is presented in Fig. 3. Similar flow-charts as presented in the Fig. 3 were previously presented by e.g. Petersen et al. (2008), Šarlah (2008) and Liu and Yu (2011). Since all the published flow-charts describe AMR model based on

the inverse Brayton cycle, no significant differences between them and the flow-chart shown in the Fig. 3 can be noted.

Since the actual values of  $c_s$  strongly depend on the current values of the solid temperature and the magnetic field, these values are applied through an iteration process. The first values of  $c_s$  at each time step are “guessed”, so the matrix equation is calculated using guessed values. The new values of  $c_s$  are further read with regard to the new calculated temperatures. This process is repeated for each time step until a particular convergence criterion is reached.

### 3. Results and discussion

The developed numerical model represents a tool for the further analysis and optimization of the AMR and its refrigeration process. In this section we will present the basic initial results in graphical form, i.e., the initial temperature response of the magnetocaloric material in the AMR, the temperature profiles during all four operational phases and the performed AMR cycle in the  $T$ - $s$  diagram. Furthermore, the impact of the basic operational parameters (the mass-flow rate and the operational frequency) and the sphere diameter of the packed-bed AMR will be presented and discussed.

#### 3.1. Basic results

The presented AMR numerical model is prepared to simulate the inverse Brayton magnetic cycle, which consists of four operational phases. For a discrete “on-off” magnetic field and fluid flow profile the initial temperature response of the hot and cold sides of the magnetocaloric material in the AMR is presented in Fig. 4. After a certain number of cycles (the number depends on the operating conditions) a steady state is achieved. This happens when the fluid temperature at the exit of the AMR in two successive cycles does not change by more than the previous described value ( $5 \times 10^{-5}$  K).

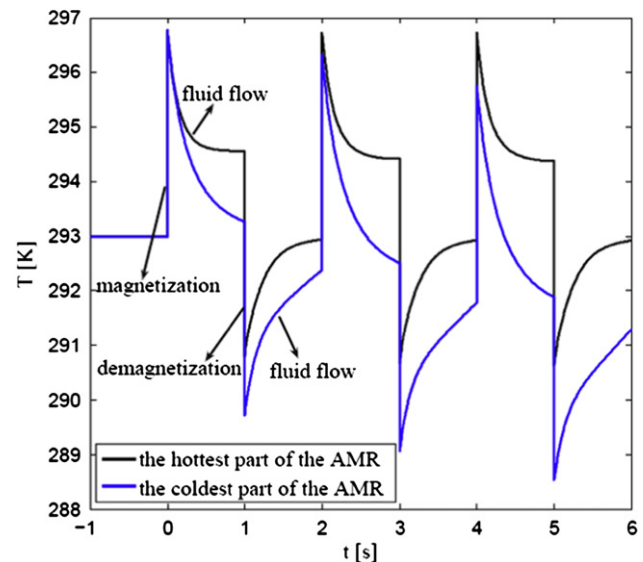


Fig. 4 – Initial temperature response of the magnetocaloric material in the AMR in the case of a discrete “on-off” magnetic field change and fluid flow profile.

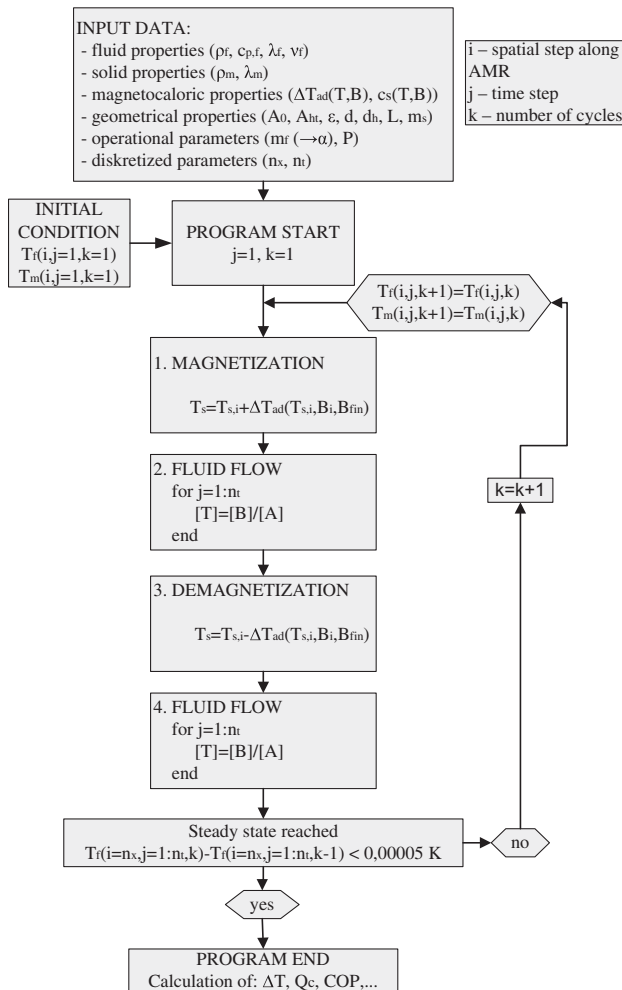


Fig. 3 – Flow-chart of the numerical model.



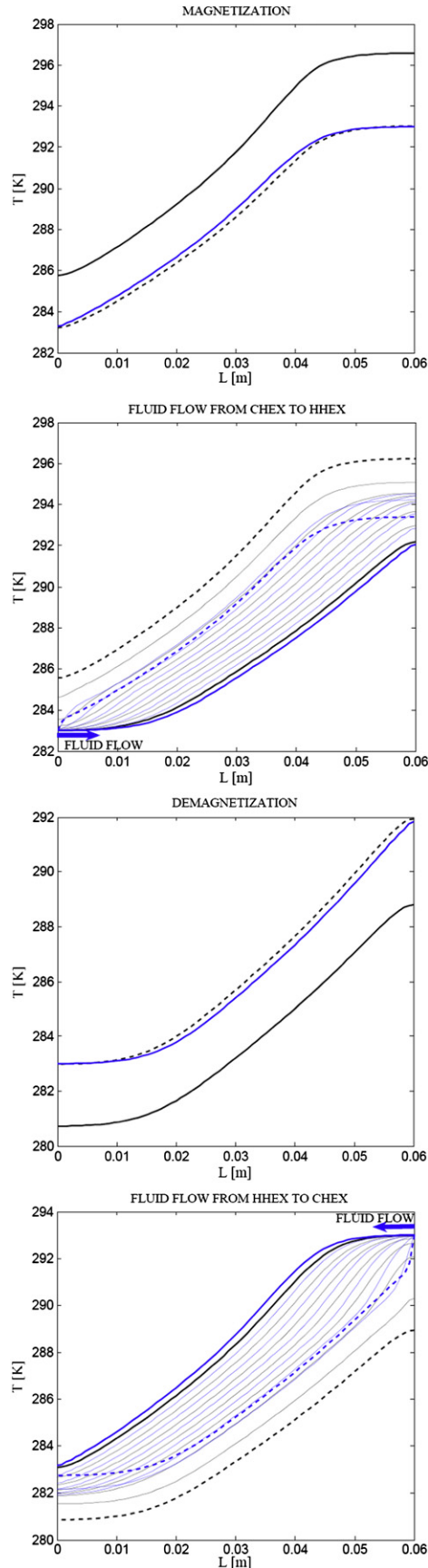


Fig. 5 – Temperature profile of the fluid (blue line) and the material (black line) along the AMR during all four

During steady state operation a temperature profile is formed along the length of the regenerator due to the counter-flow operation, where the cold and hot external heat exchangers are at different temperature levels. Fig. 5 shows the four operational phases of the AMR process for the established temperature profile. In the first phase, each part of the AMR is magnetized – its temperature increases in proportion to the MCE. In the next phase the fluid is blown from the cold to the hot heat exchanger, heating up while cooling down the material in the AMR. In the third phase, just the opposite of the first one, every part of the regenerator is demagnetized – the temperature decreases, again in proportion to the MCE. In the last phase the fluid is blown in the opposite direction, from the hot to the cold heat exchanger, completing a full cycle. It is evident from Fig. 5 that the hot part of the AMR has more pronounced temperature variation than the cold one. This is mostly due to the lower values of  $c_s$  in this temperature range and non-zero magnetic field, as previously described by Oliveira et al. (2009). According to the above description we can conclude that in the AMR process each particle in the regenerator bed performs its own inverse Brayton thermodynamic cycle, as presented in Fig. 6.

### 3.2. Optimization of the mass-flow rate and operating frequency

The operation of the AMR is affected by various independent parameters, such as the mass-flow rate, the operating frequency, the magnetic field change, the properties of the magnetocaloric material and the heat-transfer fluid, the geometry and dimensions of the AMR, the hot heat exchanger temperature, etc. In order to avoid a multi-parametric optimization the analysis was initially focused only on the optimization of the mass-flow rate and the operating frequency in a packed-bed AMR. The other parameters were considered to be constant and are presented in Table 1.

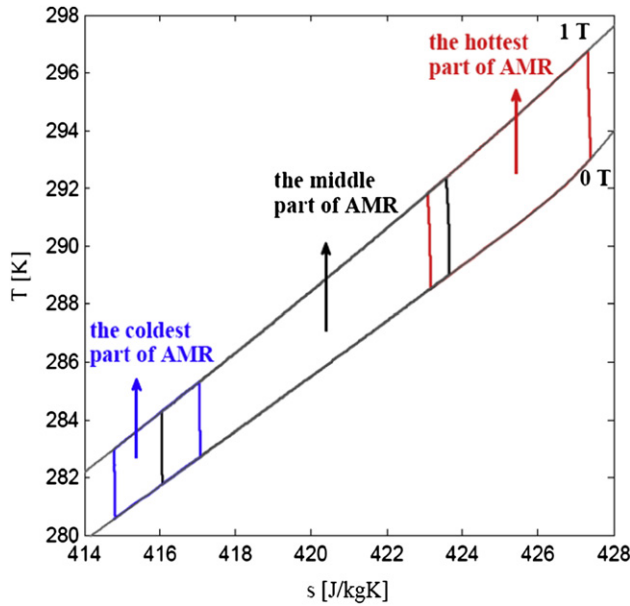
The optimization was performed for the temperature span between the fluid outlet temperatures at both ends of the AMR (under no-load conditions), for the cooling load in the cold external heat exchanger and for the COP of the AMR process. The cooling load and the COP are defined as follows:

$$\dot{Q}_c = \frac{1}{2P} \int_0^{2P} \dot{m}_f \cdot c_f \cdot (T_{CHEX} - T_f(x=L, t)) \cdot dt \quad (13)$$

$$COP = \frac{\dot{Q}_c}{(\dot{Q}_h - \dot{Q}_c)} + W_{pump} \quad (14)$$

where the difference between the thermal load ( $\dot{Q}_h$ ) and cooling load ( $\dot{Q}_c$ ) represents the applied magnetic work and the friction losses of the fluid flow. The cooling load calculated through the Eq. (13) is defined as a product of the fluid capacity

**operational phases in the steady state. The dotted lines represent the temperature profiles at the beginning of each phase, while the solid lines represent the temperature profiles at the end of each phase. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)**



**Fig. 6 – Brayton AMR thermodynamic cycle: An example for three different parts of the magnetocaloric material (the coldest, the middle and the hottest part).**

rate ( $\dot{m} \cdot c_f$ ) with the temperature difference between the cold heat exchanger temperature ( $T_{CHEX}$ ) and the fluid outlet temperature at the cold side of the AMR ( $T_f(x = L, t)$ ). This is defined during the isofield heat-transfer process and further integrated over the entire AMR cycle. The work required to pump the fluid through the AMR is defined as:

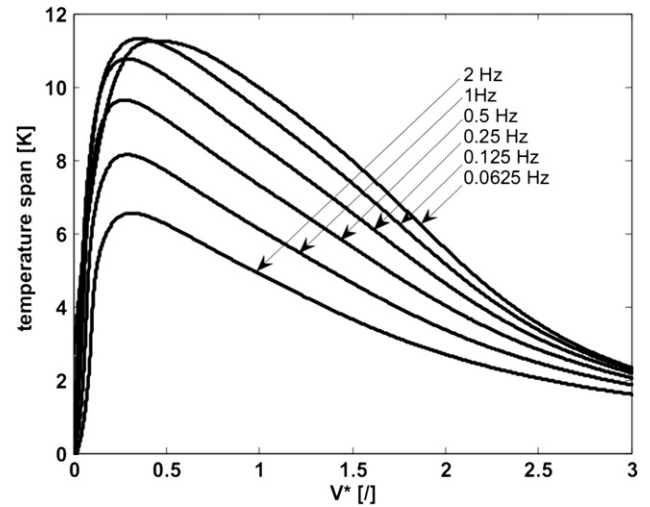
$$W_{pump} = \frac{\dot{m}_f \cdot \Delta p}{\rho_f \cdot \eta} \quad (15)$$

The analysis considered the temperature span under no-load conditions, as well as the cooling load and the COP at different temperature spans (6 K and 3 K). These two prescribed temperature spans are defined as the temperature difference between the inlet fluid temperatures from the cold and hot external heat exchanger. The results are presented for different frequencies versus the ratio of displaced fluid ( $V^*$ ). The ratio of displaced fluid  $V^*$  was introduced by [Oliviera et al. \(2009\)](#). The  $V^*$  is defined as the ratio of the fluid mass displaced during the flow period with the respect to the mass of the fluid inside the AMR:

$$V^* = \frac{\dot{m}_f \cdot P}{\rho_f \cdot V_{AMR} \cdot \varepsilon} \quad (16)$$

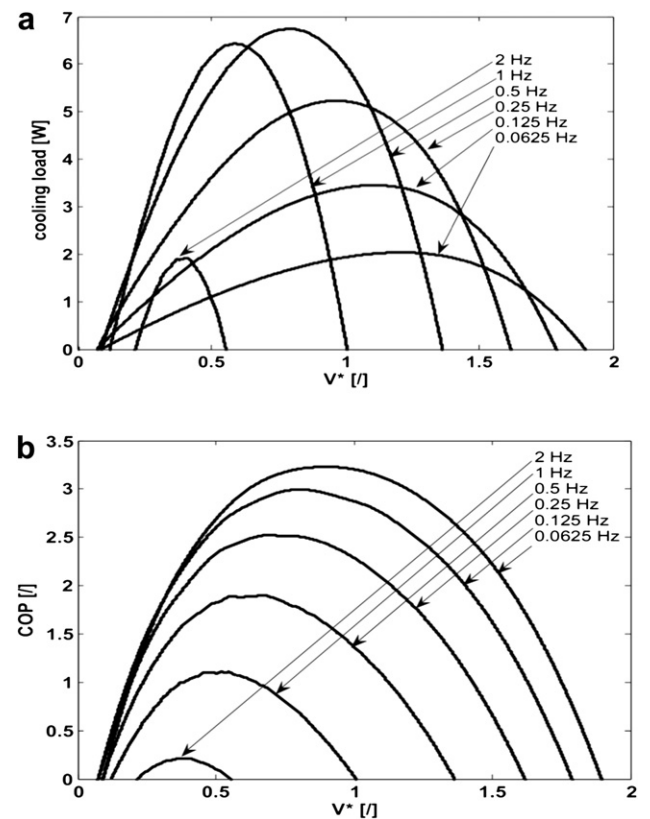
**Table 1 – Operating parameters of the AMR used in the simulations.**

|                                 |                            |
|---------------------------------|----------------------------|
| Magnetic field change           | 0.0001–1T                  |
| Magnetocaloric material         | gadolinium                 |
| Mass of magnetocaloric material | 0.185 kg                   |
| AMR dimensions                  | 0.06 m × 0.04 m × 0.016 m  |
| AMR geometry                    | spheres with 2 mm diameter |
| Heat-transfer fluid             | water                      |
| Hot heat exchanger temperature  | 20 °C                      |



**Fig. 7 – The temperature span at 0 W cooling load versus the ratio of displaced fluid mass for different operating frequencies.**

[Fig. 7](#) shows the temperature span at 0 W of cooling load. It is evident that there is an optimal ratio of displaced fluid (and consequently an optimal mass-flow rate) for each operating frequency. If the amount of displaced fluid is too small, only a part of the cooling load produced by the magnetocaloric



**Fig. 8 – The cooling load (a) and the COP (b) at 6 K of the temperature span versus the ratio of displaced fluid mass for different operating frequencies.**

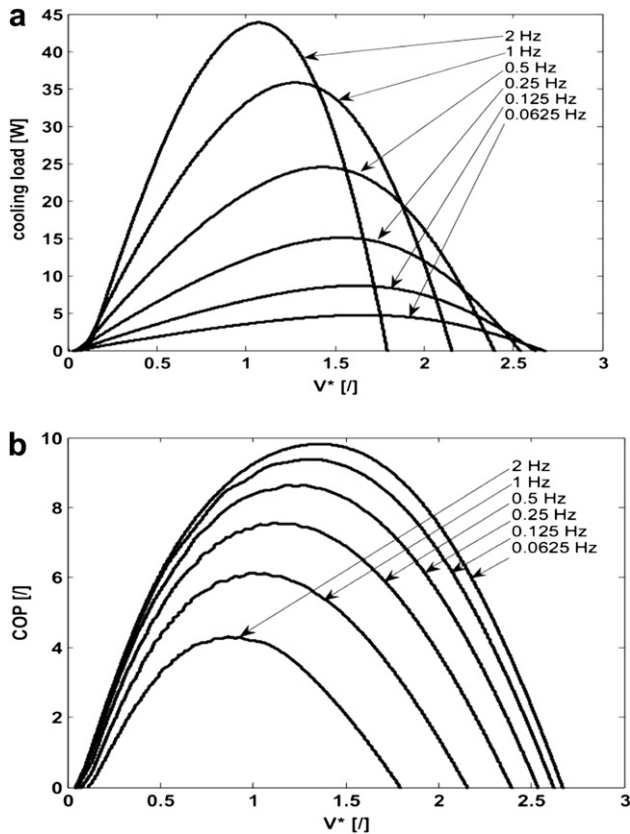


Fig. 9 – The cooling load (a) and the COP (b) at 3 K of the temperature span versus the ratio of displaced fluid mass for different operating frequencies.

effect is utilized. On the other hand, if the amount of the displaced fluid is too high, the cold reservoir will absorb heat from the hot part of the AMR or even from the hot reservoir, as explained by Petersen et al. (2008). According to Fig. 7 the temperature span increases with decreasing frequency. This is mainly due to the smaller heat-transfer irreversible losses. Note that the irreversible heat-transfer losses between the fluid and the solid increase with the frequency. This occurs because the material power density is too large to be absorbed by the fluid in such a short period of time. It is also evident that for frequencies above 0.25 Hz there is an optimal value of the ratio of the displaced fluid ( $V^* \sim 0.3$ ) at which the maximum temperature span occurs. Under these operating conditions, regardless of what the frequency is, the AMR reaches the largest temperature span when one third of the fluid in the AMR is displaced during the flow period. The results for the no-load conditions certainly do not make much sense for real applications; however, they may enhance our knowledge of the dynamic behavior and the operating characteristics of the AMR.

Figs. 8 and 9 show the cooling load and the COP at 6 K and 3 K of temperature span, respectively. It is evident that the maximum cooling load (at the optimal  $V^*$  for each frequency) in general increases with increasing frequency. However, this trend is limited by the temperature span of the AMR (Fig. 7). This is particularly so for 6 K of temperature span (Fig. 8a), where the increase in the maximum cooling loads is limited to

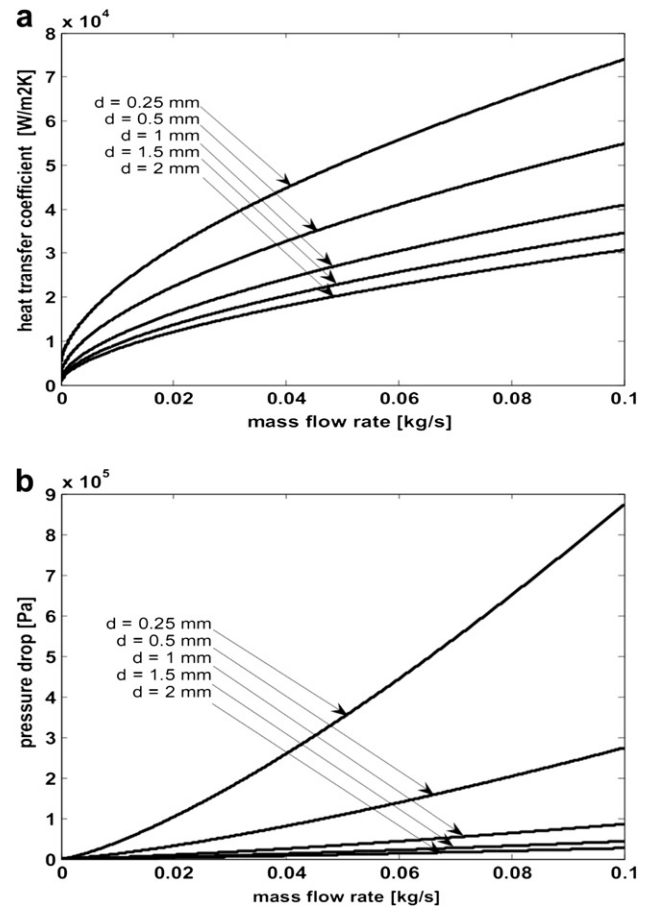


Fig. 10 – The heat-transfer coefficient (a) and the pressure drop (b) versus the mass-flow rate for different sphere diameters for the chosen parameters of the AMR (see Table 1).

frequencies of up to 0.5 Hz. A further increase in the frequency ( $>0.5$  Hz) leads to a reduction in the maximum cooling load. Under such operating conditions the AMR is no longer able to perform a sufficient temperature span (as presented in Fig. 7). For a smaller temperature span, e.g., 3 K (Fig. 9a), the increase

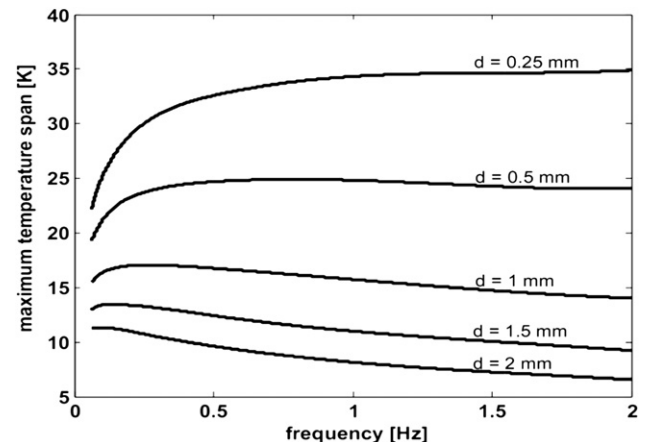


Fig. 11 – Maximum temperature span (at 0 W cooling load) for each operating frequency (at optimal  $V^*$ ) for different sphere diameters.



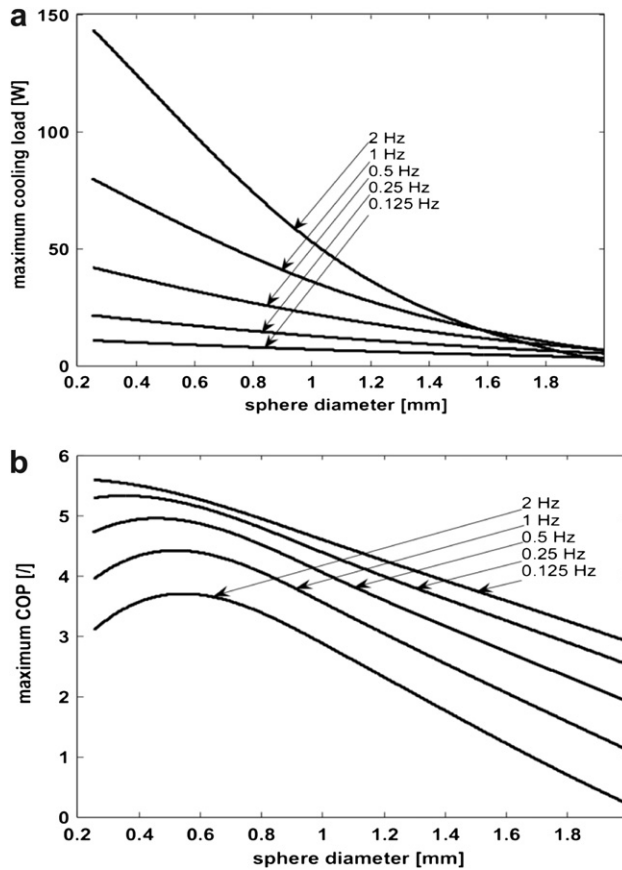


Fig. 12 – The maximum cooling load (a) and the maximum COP (b) for 6 K of temperature span versus the sphere diameter for different operating frequencies.

in the maximum cooling load is limited to frequencies slightly above 2 Hz. Figs. 8b and 9b show that the maximum COP (at the optimal  $V^*$  for each frequency) decreases with increasing frequency. This is mainly due to the fact that the optimal mass-flow rate (at which the maximum COP is obtained) strongly increases with the frequency, although the optimal ratio of the displaced fluid decreases (according to Eq. (16) and Figs. 8b and 9b). In other words, higher frequencies and, consequently, higher mass-flow rates reduce the maximum COP due to the higher viscous losses.

Similar investigations as presented in the Section 3.2 were previously presented by other authors as well. Direct comparison between results is not possible, since different authors assumed different operating conditions (e.g. magnetic field change, mass of magnetocaloric material, AMR geometry, etc.). However, the results presented in this section (Figs. 7–9) have the same trend of dependency as some of the results presented in e.g. Kawanami (2007), Bouchekara et al. (2008), Tagliafico et al. (2009), Nielsen et al. (2009) and Liu and Yu (2011).

### 3.3. Optimization of the sphere diameter

The same analysis as presented above was performed for different sphere diameters (2 mm, 1.5 mm, 1 mm, 0.5 mm and

0.25 mm). In order to obtain comparable results it was assumed that the porosity of the AMR and the mass of the magnetocaloric material do not depend on the sphere diameter and remain constant for all diameters ( $\varepsilon = 0.39$ ). This is true for the relatively small sphere diameters (a large number of particles and a uniform particle size) with respect to the frontal AMR area (Kaviani, 1995). However, the sphere diameter has a large impact on the hydraulic diameter and the heat-transfer area. Therefore, it affects the heat-transfer coefficient and the viscous losses in the AMR. Fig. 10 shows the heat-transfer coefficient and the pressure drop for different sphere diameters. It is evident that the heat-transfer coefficient and the pressure drop increase with a decreased sphere diameter. This influences the temperature span, the cooling load and the COP of the AMR. Furthermore, the sphere diameter also influences the temperature distribution inside the sphere due to the thermal conductivity of the material. Larger the sphere diameter is more pronounced the non-uniformity of the temperature distribution will be. This effect is, to the certain extend, taken into account with the application of the effective heat-transfer coefficient ( $\alpha_{eff}$ ).

Fig. 11 shows the maximum temperature span that was defined at the optimal  $V^*$  for each frequency and for different sphere diameters. It can be concluded that the AMR with the smaller sphere diameters can lead to a higher temperature span

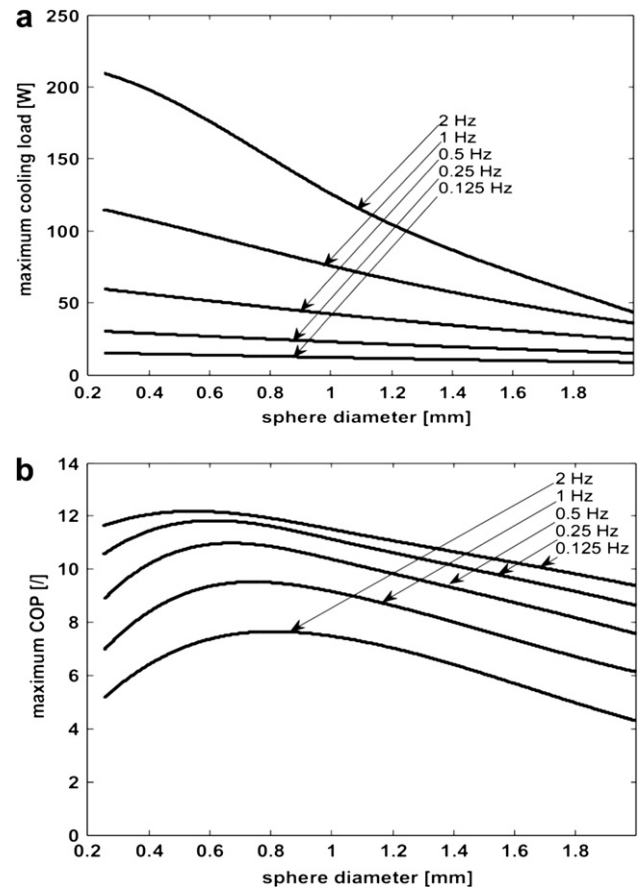
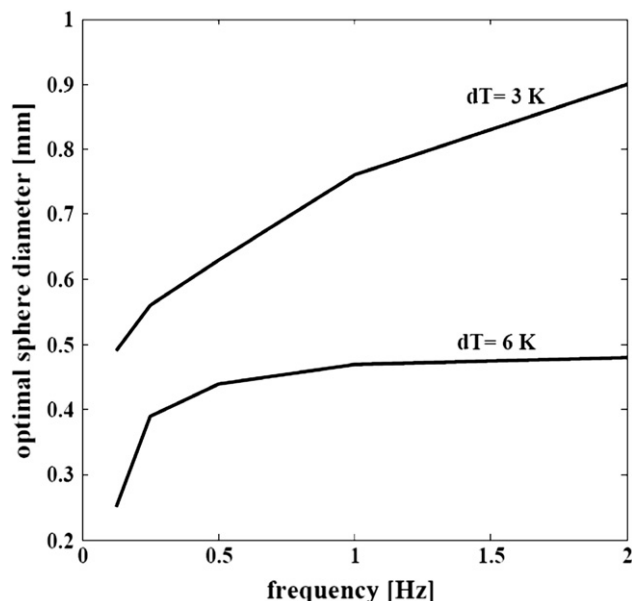


Fig. 13 – The maximum cooling load (a) and the maximum COP (b) at 3 K of temperature span versus the sphere diameter for different operating frequencies.



**Fig. 14 – The optimal sphere diameter for each frequency at which the largest maximum COP is obtained (for both analyzed temperature spans).**

due to the larger heat-transfer coefficient and the larger heat-transfer area. The largest maximum temperature span for each sphere diameter occurs at a certain operating frequency, which can be considered as optimal. The optimal operating frequency increases with a decreasing sphere diameter. This trend is strongly related to the heat-transfer irreversible losses between the fluid and the solid. As mentioned above, the heat-transfer irreversible losses increase with the frequency. Therefore, a higher frequency leads to smaller maximum temperature spans. This is especially evident for the AMR with the larger sphere diameters. However, for the AMR with the smaller sphere diameters the maximum temperature span increases with the frequency (for the analyzed frequency range). This is because smaller spheres substantially improve the heat-transfer characteristics and thus reduce the heat-transfer irreversible losses. We can conclude that the AMR with the smaller sphere diameters can generate a higher cooling load even at higher temperature spans (since the cooling load in general increases with an increasing frequency).

Figs. 12 and 13 show the maximum cooling load and the maximum COP for each frequency (at the optimal  $V^*$  with regard to the cooling load and the COP, respectively). The maximum cooling load and the maximum COP are shown for different sphere diameters and two different temperature spans (6 K and 3 K). According to Figs. 12 and 13, the maximum cooling load increases with the increasing frequency, while the maximum COP decreases. It is further evident that the maximum cooling load increases with a decreasing sphere diameter for non-zero temperature spans. In the case of the maximum COP the trends are not so straightforward, since a smaller sphere diameter not only improves the heat-transfer in the AMR, but also increases the viscous losses. We can conclude that the maximum COP increases with a reduced sphere diameter for relatively large sphere diameters.

However, the largest maximum COP is obtained at a certain optimal ratio between the heat-transfer properties and the viscous losses. Because of this, an optimal sphere diameter can be defined for each temperature span and for each frequency. A further decrease in the sphere diameter leads to increased viscous losses and causes a decrease in the maximum COP.

#### 4. Conclusions

A 1D numerical model for a reciprocating AMR has been developed. The model was applied to analyze a wide range of operating conditions (the mass-flow rate and the operating frequency) and the Gd sphere diameters of a packed-bed AMR. The following conclusions can be drawn:

- At each frequency there is a mass-flow rate that leads to the best performance of the AMR. The higher is the frequency, the higher the optimal mass-flow rate will be.
- The maximum cooling load is obtained at a higher frequency compared to the maximum COP. This is mostly due to the higher viscous losses.
- Better heat-transfer properties in the AMR with a smaller sphere diameter reduce the heat-transfer irreversible losses for lower operating frequencies (<2 Hz).
- A smaller sphere diameter leads to a larger cooling load due to the better heat-transfer properties. The higher is the temperature span, the larger the impact of the sphere diameter will be. At a zero temperature span the cooling load is, in general, independent of the sphere diameter.
- For each frequency the largest maximum COP is obtained with different sphere diameters. The lower is the frequency, the smaller the optimal sphere diameter will be (Fig. 14). The reason for this is the fact that the optimal mass-flow rate and, consequently, the viscous losses increase with the frequency. At higher frequencies the viscous losses dominate more at larger sphere diameters than at lower frequencies.
- The higher is the temperature span, the smaller the optimal sphere diameter will be (Fig. 14). At a higher temperature span the maximum cooling load (Figs. 12a and 13a) increases more rapidly with a decreasing sphere diameter than at a smaller temperature span. Thus, at a higher temperature span the viscous losses dominate at smaller sphere diameters than at a smaller temperature span.

Future work will involve the implementation and validation of the numerical model by the AMR experimental setup and the prototype of the magnetic refrigerator (which is currently in its testing phase).

#### REFERENCES

- Bouchard, J., Nesreddine, H., Galanis, N., 2009. Model of a porous regenerator used for magnetic refrigeration at room temperature. *Int. J. Heat Mass Transf.* 52 (5–6), 1223–1229.
- Boucekara, H., Kedous-Lebouc, A., Dupuis, C., Allab, F., 2008. Prediction and optimization of geometrical properties of the refrigerant bed in an AMRR cycle. *Int. J. Refrigeration* 31 (7), 1224–1230.

- Dankov, S.Y., Tishin, A.M., Pecharsky, V.K., Gschneidner Jr., K.A., 1998. Magnetic phase transition and the magnetothermal properties of gadolinium. *Phys. Rev. B* 57 (6), 3478–3490.
- Egolf, P.W., Manz, H., 1994. Theory and modeling of phase change materials with and without mushy regions. *Int. J. Heat Mass Transf.* 37 (18), 2917–2924.
- Engelbrecht, K., Nellis, G., Klein, S., 2005. A Numerical Model of an Active Magnetic Regenerator Refrigeration System. Final Report. Univ. Wisconsin-Madison.
- Hashimoto, T., Numasawa, T., Shino, M., Okada, T., 1981. Magnetic refrigeration in temperature range from 10 K to room temperature: the ferromagnetic refrigerants. *Cryog.* 21 (11), 647–653.
- Kaviany, M., 1995. *Principles of Heat Transfer in Porous Media*, second ed. Springer.
- Kawanami, T., 2007. Heat transfer characteristics and cooling performance of an active magnetic regenerator. In: *Proc. 2nd Int. Conf. Magn. Refrigeration Room Temp., IIF/IIR*, 23–34.
- Kays, W.M., London, A.L., 1984. *Compact Heat Exchangers*. McGraw-Hill Book Company, New York.
- Liu, M., Yu, B., 2011. Numerical investigations on internal temperature distribution and refrigeration performance of reciprocating active magnetic regenerator of room temperature magnetic refrigeration. *Int. J. Refrigeration* 34 (3), 617–627.
- Nielsen, K.K., Bahl, C.R.H., Smith, A., Bjoerk, R., Pryds, N., Hattel, J., 2009. Detailed numerical modeling of a linear parallel plate active magnetic regenerator. *Int. J. Refrigeration* 32 (6), 1478–1486.
- Nielsen, K.K., Tušek, J., Engelbrecht, K., Schopfer, S., Kitanovski, A., Bahl, C.R.H., Smith, A., Pryds, N., Poredoš, A., 2011. Review on numerical modeling of active magnetic regenerators for room temperature applications. *Int. J. Refrigeration* 34 (3), 603–616.
- Oliviera, P.A., Trevizioli, P.V., Barbosa, J.R., Prata, A.T., 2009. Numerical analysis of a reciprocative active magnetic regenerator. In: *Proc. 3rd Int. Conf. Magn. Refrigeration Room Temp., IIF/IIR*, 283–294.
- Petersen, F.P., Pryds, N., Smith, A., Hattel, J., Schmidt, H., Knudsen, H.J.H., 2008. Two-dimensional mathematical model of a reciprocating room-temperature active magnetic regenerator. *Int. J. Refrigeration* 31 (3), 432–443.
- Sarlah, A., 2008. Thermo-hydraulic properties of heat regenerators in magnetic refrigerator. PhD thesis, University of Ljubljana.
- Shah, R.K., Sekulic, D.P., 2003. *Fundamentals of Heat Exchanger Design*. John Wiley & Sons, Inc., Hoboken, New Jersey.
- Shir, F., Mavriplis, C., Bennett, L.B., Torre, E.D., 2005. Analysis of room temperature magnetic regenerative refrigeration. *Int. J. Refrigeration* 28 (4), 616–627.
- Siddikov, B.M., Wade, B.A., Schultz, D.H., 2005. Numerical simulation of the active magnetic regenerator. *Comput. Math. Appl.* 48 (9–10), 1525–1538.
- Smaili, A., Chahine, R., 1998. Thermodynamic investigations of optimum active magnetic regenerators. *Cryog* 38 (2), 247–252.
- Tagliafico, G., Scarpa, F., Tagliafico, L.A., 2009. Dynamic 1D model of an active magnetic regenerator: a parametric investigation. In: *Proc. 3rd Int. Conf. Magn. Refrigeration Room Temp., IIF/IIR*, 245–50.
- Wakao, N., KEGUI, S., 1982. *Heat and Mass Transfer in Packed Beds*. Routledge.